Course: CSE 409 (Digital Signal Processing Lab)

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DSP Lab Problems (MATLAB)

Problem 1: Basic Signal Generation

Generate and plot the following signals:

- Unit step u[n]
- Unit impulse δ[n]
- Ramp signal
- Sine wave
- Cosine wave

  Plot them in separate subplots and explain differences.

Problem 2: Signal Operations

Given x[n]={1,2,3,4}:

- Perform time shifting, scaling, and reversal
- Plot all results
  Verify mathematically.

Problem 3: Even and Odd Decomposition

Take any discrete signal and:

- Decompose into even and odd components
- Verify: x[n]=xe[n]+xo[n]

Problem 4: Sampling of Analog Signal

Sample a sine wave at:

- Nyquist rate
- Above Nyquist
- Below Nyquist
  Plot and compare.

Problem 5: Aliasing Demonstration

Generate a continuous-time sinusoid and:

- Sample at different frequencies
- Show aliasing effect visually

Problem 6: Reconstruction

- Sample a signal
- Reconstruct using interpolation (interp1)
  Compare original vs reconstructed.

Problem 7: Uniform Quantizer

- Implement a uniform quantizer
- Quantize a sine wave
- Plot input vs quantized output

Problem 8: Quantization Error Analysis

- Compute quantization error
- Plot error signal
- Calculate Mean Squared Error (MSE)

Problem 9: Effect of Bit Resolution

- Quantize signal using 2, 4, 8 bits
- Compare signal quality and error

Problem 10: Discrete Fourier Transform (DFT)

- Compute DFT using:
  - Built-in fft
  - Manual implementation
    Compare results

Problem 11: Frequency Spectrum Analysis

- Generate composite signal (multiple sinusoids)
- Plot magnitude spectrum
  Identify frequency components

Problem 12: Short-Time Fourier Transform (STFT)

- Apply spectrogram() on:
    - Stationary signal
    - Non-stationary signal
      Interpret time-frequency behavior

Problem 13: Statistical Measures

For a given signal:

- Mean
- Variance
- Standard deviation
- Skewness
- Kurtosis
  Use MATLAB built-ins and manual formulas

Problem 14: Correlation & Covariance

- Compute:
    - Auto-correlation
    - Cross-correlation
    - Covariance
      Interpret physical meaning

Problem 15: Random Signal Analysis

- Generate white Gaussian noise
- Analyze statistical properties
  Verify theoretical expectations

Problem 16: Low-Pass Filter Design

- Design FIR low-pass filter using fir1
- Apply on noisy signal
  Show before/after filtering

Problem 17: High-Pass Filter Design

- Remove DC component from signal
  Compare original and filtered

Problem 18: IIR Filter Design

- Design Butterworth filter using butter
- Apply filtering using filter

Problem 19: Frequency Response

- Plot magnitude and phase response using freqz
  Interpret cutoff frequency

Problem 20: Filter Comparison

- Compare FIR vs IIR:
  - Response
  - Stability
  - Phase characteristics

Problem 21: Noise Removal Project

- Add noise to signal
- Apply appropriate filter
  Evaluate performance using SNR

Problem 22: Speech Signal Analysis (Optional Advanced)

- Load audio file (audioread)
- Apply:
  - STFT
  - Filtering
    Observe real-world signal behavior

Problem 23: Aliasing + Filtering Combined

- Create aliased signal
- Apply anti-aliasing filter
  Analyze improvement

Problem 24: Signal Classification (Basic Intro)

- Generate multiple signal types
- Extract features (mean, variance, etc.)
  Classify based on properties